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Abnormal Returns in Time of Crisis: An Event Study of Norwegian and European Stocks during the Corona Pandemic

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Abstract

This paper investigates whether the world health organization (WHO) announcement about declaring COVID-19 a world concerning pandemic, generated abnormal returns for Norwegian Air Shuttle NAS, as measured against expected returns from the Capital Asset Pricing Model (CAPM). At the same time a broad European multi-firm analysis was carried out to improve the robustness and validity of our results, by using 9 additional European aviation firms. The research question is to what extent did the stock prices of aviation firms incorporate new publicly available information related to the corona pandemic?

Taken together, the evidence is most consistent with the semi-strong form of market efficiency, with frictions. The multi-firm results show a broad, value relevant reaction to the WHO announcement, indicating that public information was incorporated into prices. The single-firm results for NAS are more volatile, mostly negative but less precisely estimated, where full repricing required several days. Which is discussed to be caused by confounding events, illiquidity, firm-specific uncertainty, overlapping domestic policy news, and limits to arbitrage that slow price discovery. Thus, the findings go beyond weak-form efficiency but fall short of strong-form immediacy, best described as semi-strong efficiency with delayed information under high uncertainty.

Overall, the evidence points to semi-strong efficiency with frictions, where public news was incorporated, but the pandemic's cash-flow shock required several days to be fully priced, yielding a broad negative effect and a noisier and slower adjustment.

Introduction

The year 2020 had one of the most severe global economic shocks in modern history, this was not something we have earlier familiarity with. The corona-pandemic was a rapid spreading virus, and the subsequent policy responses were devastating to economic markets with reliance on face-to-face interactions, some examples of severely affected industries are the tourist industry, traveling industry and retail & services industries. Reasons behind the negative shock was the uncertainty and restrictions around the pandemic situation, where global travel restrictions, border closures and mandatory quarantines led to a collapse in consumer spending. At the same time there was a high number of general uncertainties in financial markets (Altig et al 2020), an example of uncertainty and unrest was that the S&P 500 and S&P Europe 350, who lost more than a third of their value under the corona pandemic relative to their historical maximum (Martins & Cro 2022). All industries got affected by the pandemic to at least some extent, but the industry of aviation services was fundamentally unprotected against major restrictions and shifts in general aspects of the market.

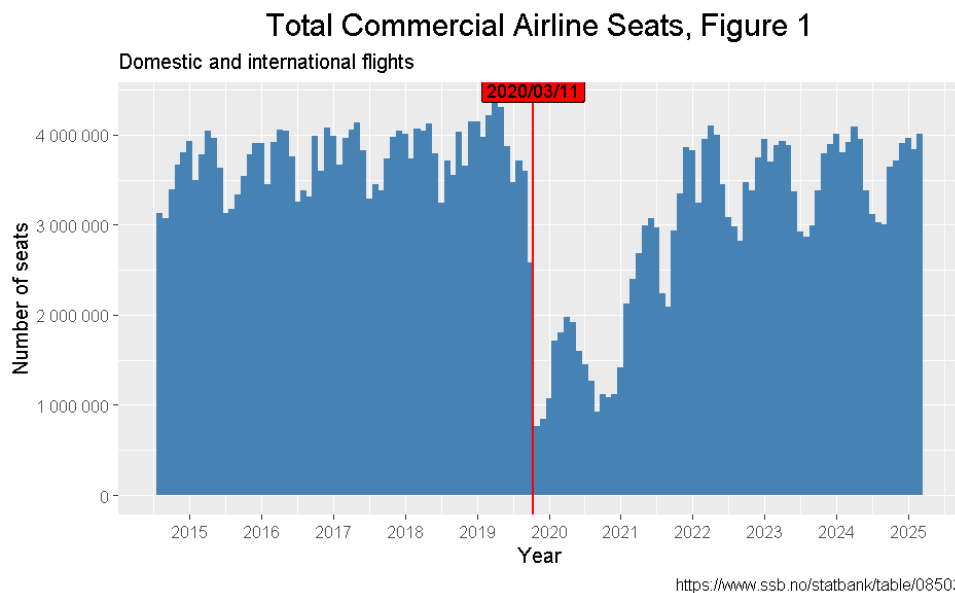


Figure 1 shows the total Norwegian commercial airline seats over time, which is made by us with RStudio and data from SSB (2025). The quantity of commercial airline seats represents the demand for travel by air where the fluctuations correspond to standard seasonal demand cycles.

Figure 1 shows us the major drop in total number of seats around the time of the corona pandemic, the red indicator shows the date 11th March 2020. This is the day where *World Health Organization* (WHO 2020) declared the corona virus as a global pandemic, which led to a mayor negative demand shock for airline seats and changes in revenue expectations and liquidity shortages. Such key announcements are shown to give negative cumulative abnormal returns (CARs) in airline stocks (Maneenop & Kotcharin 2020), which translate to a lower return than the normal return.

In this study, we are analyzing the Capital Asset Pricing Model (CAPM) for Norwegian Air Shuttle (NAS), as it represents a particularly undocumented sector of aviation markets, namely the Scandinavian markets. At the same time, we are comparing the single case of NAS with a broad analysis for 9 more European airline firms. The chosen event is the WHO announcement 11th March 2020, several announcements were made around that time, both international and domestic, therefore a general announcement from WHO is interpreted as a summary of the events.

The hypothesis is developed from the *efficient market hypothesis* (Fama 1970), specifically its semi-strong form. According to the market hypothesis, asset prices fully reflect all publicly available information, implying that any new event, such as WHO's announcement, should be immediately incorporated into the firm's stock price. In an efficient market, abnormal returns would only occur at the exact time of the information release and would not persist beyond the short-term event window. By analyzing Norwegian and European stock price reactions to the announcement we test whether the market efficiently processed new information, or whether delayed adjustments and prolonged abnormal returns indicate inefficiencies or overreactions in the pricing of airline equities.

To what extent did the stock prices of aviation firms incorporate new publicly available information related to the corona pandemic?

This event study establishes whether the aviation firm's stock returns around the declaration of the corona pandemic systematically differs from predictions by the CAPM model. At the same time the analysis is based upon the efficient market hypothesis from the article *Efficient Capital Markets: A Review of Theory and Empirical Work*, by Eugene F. Fama (1970), where

we test hypothesis concerning whether the stock market is fully reflecting all the information available.

The remainder of this paper is structured as follows. The first section, *Earlier existing literature*, presents a review of previous empirical research on airline stocks and market reactions during the COVID-19 pandemic. The next section, titled *The efficient market hypothesis*, provides the theoretical framework, including the Efficient Market Hypothesis (EMH), the market model, and the CAPM. The following section, *Method & Data Structure*, describes the data, sample selection, and the event-study methodology. The subsequent section, *Analysis & Discussion*, presents the empirical results, and the final section concludes the paper

Earlier existing literature

Previous studies have primarily focused on the global stock market response to the COVID-19 outbreak rather than national cases such as Norway. Maneenop and Kotcharin (2020) examine the global airline industry and document large negative abnormal returns following the World Health Organization's pandemic declaration on March 11, 2020. The effect was strongest in Western markets, including Australia, the United States, the United Kingdom, and Canada, where airline stocks were among the weakest post-event performers. Martins and Cró (2022) extend this literature by analyzing how vaccine announcements affected airline stocks. They found that the WHO declaration caused significant negative CAR, whereas announcements of vaccine effectiveness generated positive market reactions, particularly for the Pfizer–BioNTech vaccine. Together, these findings indicate that investors reacted negatively to market uncertainty but positively to news that could stabilize or positively affect the market.

Martins and Cró (2022) also found that airline stock reactions were reinforced or mitigated by several firm-specific characteristics. Some of these factors are relevant to our analysis of NAS stock prices, where liquidity, size, leverage and business models can be used to explain the direction of the reactions. Tretheway and Markhvida (2014) find that the demand for airline goods is highly income elastic, where a general recession will most likely affect the total demand with a rapid fall. For our situation we can then expect a major reaction to the corona pandemic announcement.

However, the existing literature focuses on large international samples. Evidence from the Norwegian market, and specifically NAS, remains limited. This study aims to fill that gap by examining how investors reacted to the domestic restrictions announced on March 11, 2020.

In this study we are using the efficient market hypothesis (Fama 1970), market model (MacKinlay 1997), CAPM (Sharpe 1964), abnormal return, cumulative abnormal return and their respective average forms (MacKinlay 1997, Xue 2025), which will be shown in the next chapters. One reasoning worth noticing is the ceased use of these methods in newer studies, MacKinlay (1997) discusses that the validity of the restrictions imposed by the CAPM on the market model is questionable, Fama & French (1996) explains this problematic conclusion. A more relevant method is by expanding the market model and the CAPM with other relevant factors such as momentum, size and other value factors. Such models require much more substantial analysis. For our case, with the time and page limit, we have decided to go with the simpler method by using the models described above.

The efficient market hypothesis

To evaluate whether markets react efficiently to new information, it is necessary to define what “efficiency” means in economic terms. The Efficient Market Hypothesis developed by Fama (1970), provides such a framework by linking information flow to price adjustments.

Financial markets allocate the economy’s capital efficiently by ensuring that prices reflect available information. In an ideal setting, stock prices provide accurate signals for resource allocation, allowing investors to make decisions under the assumption that prices fully reflect all available information. A market in which prices always fully reflect available information is defined as *efficient* (Fama, 1970). In such a market, firms can make investment decisions and investors can allocate their portfolios based on prices that represent all known information. Any deviation from this efficiency would imply that some information is not yet incorporated into prices, creating potential opportunities for abnormal returns.

According to Fama (1970), there are three forms of an efficient market: weak, semi – strong and strong form.

Weak form efficiency implies that current stock prices fully reflect all information contained in past prices. Consequently, trading strategies based on historical patterns such as technical analysis should not generate consistent abnormal returns.

Semi-strong form efficiency extends this idea by stating that prices adjust rapidly to all publicly available information, including earnings announcements, macroeconomic data, and policy changes. Under this form, neither technical nor fundamental analysis should provide excess returns.

Strong form efficiency represents the highest level, where prices fully incorporate all information public and private alike. In such a market, even investors with insider information would be unable to earn abnormal profits.

The market model and abnormal return

This study focuses on the semi-strong form of efficiency, as it tests how quickly Norwegian Air Shuttle's stock price incorporated publicly available information about the COVID-19 pandemic. To empirically test these predictions, we apply the event study methodology of MacKinlay (1997), the market model. This approach quantifies how a specific event influences firm value by comparing actual stock returns with a model of expected returns in the absence of the event. MacKinlay (1997) presents the market model as

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t} \quad (1)$$

where $R_{i,t}$ and $R_{m,t}$ represents the period returns on securities and market portfolio, where i denotes security, t denotes period, m denotes market portfolio. $\varepsilon_{i,t}$ is the term for zero mean disturbance. This model is estimated by the ordinary least square method (OLS), which is shown to be a consistent method for the market model parameters (MacKinlay (1997)). The market models rely on some assumptions which is based on multivariate normal asset returns, where the returns should be independently and identically distributed through time (MacKinlay (1997)). MacKinlay (1997) explains that these assumptions are relatively strong but shows to be empirically reasonable and robust in practice.

$$E[\varepsilon_{i,t} | R_{m,t}] = 0, \quad Var(\varepsilon_{i,t}) = \sigma_\varepsilon^2 \quad (2)$$

The parameters of the market model, α_i , β_i , and σ_ε^2 , are the fundamentals for estimating the event effects. The model compares returns of a security to the market portfolio's returns, where an economy wide stock index is used as the market portfolio. The model specifications reduce the variance of the abnormal returns by excluding the share of returns related to the market's return variation.

The event's effect needs a measure of the abnormal return. This is the realized return of the security during the event window minus the return we would normally expect for that period if the event had not occurred. Where the normal return is estimated by the earlier explained market model. The abnormal return (AR) for firm i on day τ is defined as:

$$AR_{i,\tau} = R_{i,\tau} - E[R_{i,\tau}|X_\tau] \quad (3)$$

$$E[R_{i,\tau}|X_\tau] = \hat{\alpha}_i + \hat{\beta}_i R_{m,\tau} \quad (4)$$

where $R_{i,\tau}$ is the observed return and $E[R_{i,\tau}|X_\tau]$ represents the expected return conditional on available information, i.e. normal return, which is formed from the estimates in the market model. In a market model approach will the conditional information, X_τ , be interpreted as the market return, where we assume a linear relationship between the security return, $R_{i,\tau}$, and the market return (MacKinlay 1997). A statistically significant abnormal return indicates that the event affected the stock price beyond normal market fluctuations.

The capital asset pricing model (CAPM)

The abnormal return presented by the market model needs a theoretical fundament, this is done by presenting the capital asset pricing model (CAPM) (Sharpe 1964). The CAPM provides a theoretical relationship between expected return and systematic market risk. Let $P_{j,t}$ be the price of a stock j , at time t , the calculated one-period return is therefore:

$$R_{j,t} = \frac{P_{j,t} - P_{j,t-1}}{P_{j,t-1}} = \frac{P_{j,t}}{P_{j,t-1}} - 1 \quad (5)$$

When the stock is experiencing a market shock from an event, we denote the daily abnormal return of the stock as follows:

$$AR_{j,t} = R_{j,t} - E[R_{j,t}|Event = 0] \quad (6)$$

Which shows a difference between the actual return of the stock of the event day and the expected return of the counterfactual (Xue 2025), this is the same conclusion as for the market model. The CAPM model is taking the risk-free rate, $r_{f,t}$, into account:

$$E[R_{j,t}|Event = 0] = r_{f,t} + \beta_j(R_{m,t} - r_{f,t}) \quad (7)$$

The abnormal return will therefore be presented as:

$$AR_{j,t} = R_{j,t} - (r_{f,t} + \beta_j(R_{m,t} - r_{f,t})) \quad (8)$$

where β_i , measures the stock's sensitivity to market movements. The CAPM thus serves as a benchmark for "normal" performance. Any deviation from this benchmark, captured by the AR, represents the market's reaction to new information. In the market model we presented a regression model, the same needs to be done in the CAPM, this is the final model.

$$R_{it} - r_{f,t} = \alpha_i + \beta_i(R_{m,t} - r_{f,t}) + \varepsilon_{i,t} \quad (9)$$

The regression model is known as excess return regression CAPM, where the risk-free rate $r_{f,t}$ is taken into account by subtracting it from the expected return. Since we cannot observe expected returns in practice, we replace them with realized returns and therefore subtract the risk-free rate to isolate the risk related component of the return. This way, the intercept becomes the alpha, which measures whether the stock has outperformed or underperformed the market on a risk-adjusted basis.

To capture the overall effect of the event over time, abnormal returns are aggregated across an event window $[a, b]$:

$$CAR_{i,t}^{[a,b]} = \sum_{\tau=t+a}^{t+b} (R_{i\tau} - E[R_{i,\tau}|Event = 0]) \quad (10)$$

Here, $t = 0$ denotes the event day, while negative and positive values capture days before and after the event. A positive (negative) CAR or CAAR indicates that the event increased (decreased) firm value. If the market is efficient in the semi-strong sense, cumulative abnormal returns should not systematically differ from zero.

While the cumulative abnormal return captures the total impact of the event for a single firm, it is often useful to examine whether this effect holds on average across firms. To do so, we need the average abnormal return, (AAR), where we consider a sample of N securities. Then the cross-sectional mean abnormal return for a period is (Xue 2025):

$$AAR_t = \frac{1}{N} \sum_{i=1}^N AR_{i,t} \quad (11)$$

The individual CARs are aggregated into a Cumulative Average Abnormal Return, which summarizes the market's overall reaction to the event.

$$CAAR(a, b) = \frac{1}{N} \sum_{i=1}^N \sum_{t=a}^b AR_{it} \quad (12)$$

In this expression, the inner summation aggregates abnormal returns for firm i over the event window $[a, b]$, producing the cumulative abnormal return. The outer summation then averages these CARs across N firms, yielding the Cumulative Average Abnormal Return or ($CAAR$). This provides the mean market-wide response to the event (Xue 2025).

Hypothesis development and objectives

This study aims to test whether the WHO's announcement of the COVID-19 pandemic on March 11, 2020, had a value - relevant impact on NAS's stock price.

According to the semi-strong form of the Efficient Market Hypothesis (Fama, 1970), stock prices should fully and immediately reflect all publicly available information. Any abnormal return observed around the event therefore represents the market's reaction to new information.

Following the standard event-study framework by MacKinlay (1997), cumulative abnormal returns (CAR) are obtained by summing the daily abnormal returns within the event window. This allows us to examine whether the event led to a statistically significant deviation from expected returns.

If markets are efficient, the CAR should not differ systematically from zero, since the information would already be incorporated into prices at the time of the announcement. The hypotheses are stated as:

$$\begin{aligned} H_0: CAR_{i,[a,b]} &= 0 \\ H_1: CAR_{i,[a,b]} &\neq 0 \end{aligned} \tag{13}$$

Rejecting H_0 indicates that the WHO announcement contained new information not previously reflected in NAS's stock price, implying a deviation from semi-strong efficiency. Failing to reject H_0 suggests that prices adjusted efficiently in line with market-efficiency predictions.

While the hypotheses above apply to the single-firm case, the same logic extends to the European based multi-firm analysis. When multiple firms are examined, the null and alternative hypotheses are reformulated as:

$$\begin{aligned} H_0: CAAR_{[a,b]} &= 0 \\ H_1: CAAR_{[a,b]} &\neq 0 \end{aligned} \tag{14}$$

Testing whether the CAAR across firms differs significantly from zero. Rejecting H_0 would imply that, on average, the event had a significant impact on the stock prices of the affected firms, indicating a deviation from semi-strong market efficiency at the aggregate level.

To evaluate whether the estimated abnormal returns differ statistically from zero, a two-sided t-test is employed. For a single firm, the test examines whether the cumulative abnormal

return (CAR) over the event window differs significantly from zero. The corresponding t-statistic is computed as:

$$t = \frac{CAR_{i,[a,b]}}{\frac{s(CAR_{i,[a,b]})}{\sqrt{N}}} \quad (15)$$

For the European based multi-firm case, we test whether the Cumulative Average Abnormal Return (CAAR) differs significantly from zero. The corresponding t-statistic is:

$$t = \frac{CAAR_{[a,b]}}{s(CAAR_{[a,b]})/\sqrt{N}} \quad (16)$$

where s denotes the sample standard deviation and N the number of observations. A statistically significant t-value indicates that the event generated abnormal returns beyond normal market fluctuations.

Method & Data Structure

Research design:

We assess the stock market impact of the WHO announcement on NAS stock prices and compare the results against a multi-firm sample of European airline stock prices to test for robustness. We use a standard short-term event study design with a CAPM market model for abnormal returns. The data used in the single-firm event analysis comes from daily stock prices for Norwegian Air Shuttle (NAS) and Oslo Stock Exchange Benchmark Index (OSEBX), the multi-firm analysis uses stock prices for 9 European aviation firms, and the general European Morgan Stanley Capital International index, (MSCI European index). The data was obtained from these sources; *Investing.com*, *Yahoo Finance* and *Refinitiv*.

Stock prices are used to calculate the returns of interest, while the indexes are used to calculate the market returns, see the earlier chapters for reference. For all estimations of models, we used data from the Refinitiv platform. In the CAPM, the Norwegian Overnight Weighted Average (NOWA) rate was applied as the risk-free rate. This rate is appropriate for our case since all returns are denominated in NOK and NOWA represents a short-term risk-free rate in the Norwegian market.

For the single-firm analysis, the OSEBX index served as the market benchmark, which is a natural choice given its representation of the Norwegian equity market. In the extended CAAR analysis, we used the MSCI Europe Index as the market proxy and continued to apply NOWA as the risk-free rate. A European index was chosen because all firms in the cross-sectional sample are European, ensuring consistency and comparability across the sample. Adjusted closing price is the optimal price to use in the analysis, but since adjusted prices had limited access, the normal stock price was used instead

Descriptive statistics:

The chosen period is from 01.01.2019 to 01.01.2021, where the event day is given by 13.03.2020, $t = 0$. The event window is defined as $[-10, +10]$ trading days from $t = 0$. This extended window allows for potential information leakage prior to the official announcement, as well as delayed market reactions. However, a wider window increases the risk of capturing unrelated market noise. To ensure robustness, alternative windows such as $[-3, +3]$ and $[-5, +5]$ are also used. The estimation window is chosen to be from 26.09. 2019 to 19.02. 2020, to capture the normal return without effects of the event.

Summary for time intervals

Period	Mean	Median	Min	Max	Days
Estimation window	1808.5723	1789.530	1571.86	2180.51	100
Event window	599.0300	522.710	331.28	1063.16	21
Post-event	112.0947	61.195	14.95	446.10	192
Pre-event	2104.0645	1837.645	1237.87	4889.18	288

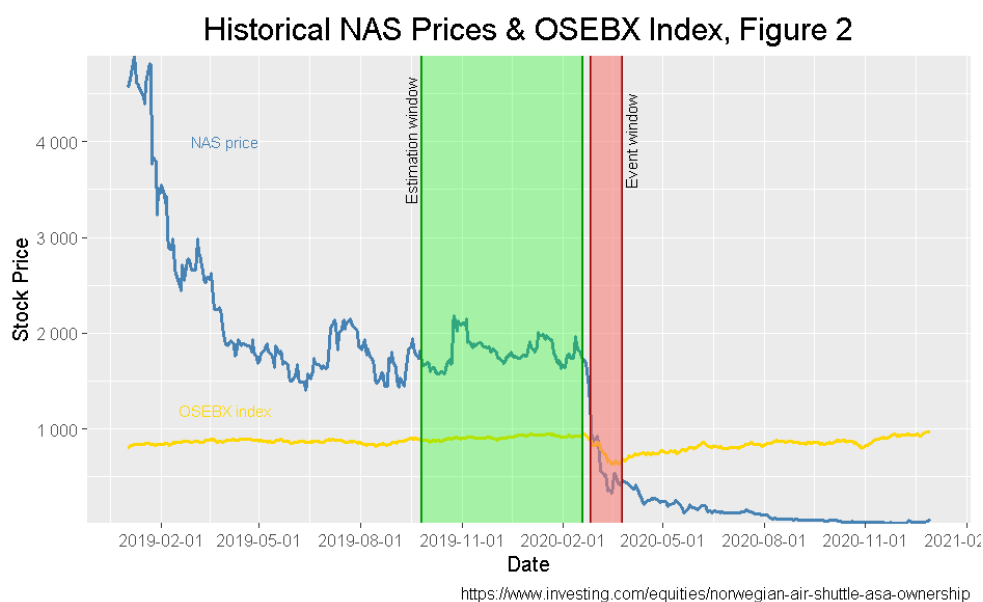
Note:

Data: <https://www.investing.com/equities/norwegian-air-shuttle-asa-ownership>

Table 1 presents the descriptive statistics of NAS stock price across four distinct time intervals; the *pre-event*, *event window*, *post-event* and the *estimation* window, the table is made in RStudio (see appendix). During the *pre-event* period, with 293 trading days before the event, the mean (2083.87) and median (1833.44) stock prices were substantially higher than the other intervals, indicating that Norwegian's stock traded at relatively high levels

before the event. In the event window, with 11 trading days, both the mean and median prices dropped drastically compared to pre-event period, with a difference from the pre-event period of -1572.1 and -1317.14 , this reflects a negative market reaction to the event. Although the event window is defined as ± 10 calendar days around the event, this corresponds to 11 trading days once weekends and non-trading days are removed. In the post-event period, with 210 trading days, mean and median price dropped to 149.14 and 99.01 , which shows a continuation of the negative market reaction after the event window, during the same period, the stock price reached a maximum stock price of 929.33 during the post-event period, indicating significant fluctuations in price after the event. The estimation window is rather stable compared to the entire pre-event window in 2019, with a mean and median of 1808.57 and 1789.53 . This suggests similar fluctuation patterns as observed in the post-event period, but with substantially higher price levels.

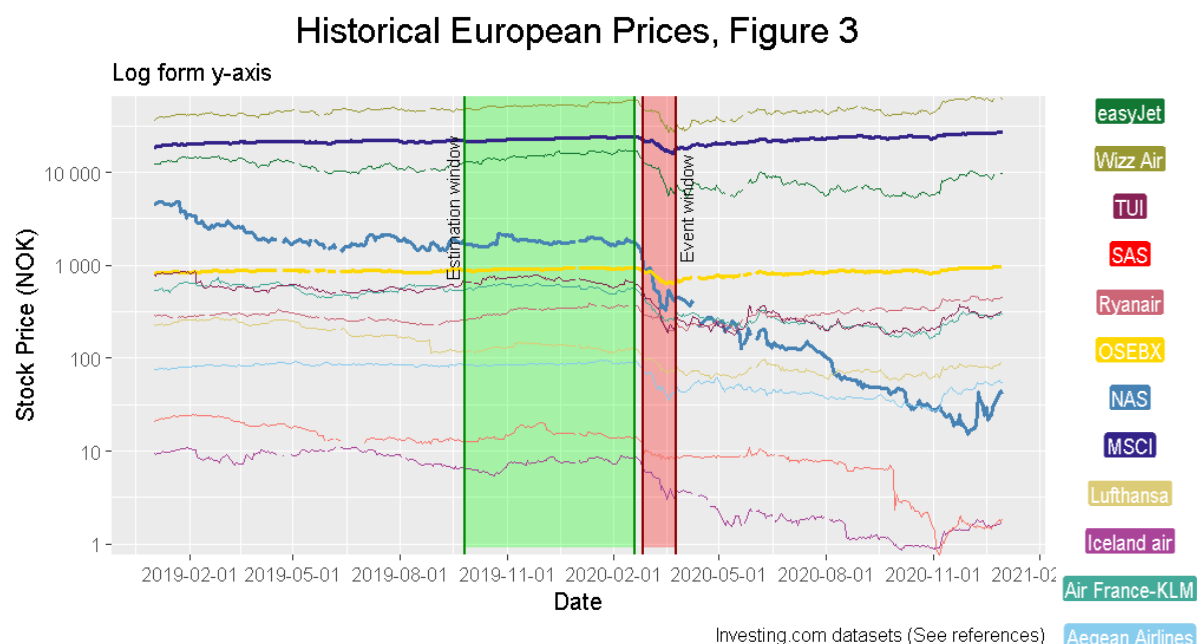
Figure 2 shows NAS stock prices and OSEBX index, We are using the index as a reference for the development of expected NAS returns, the figure shows a similar development as the NAS stock price within the event window, this shows that the entirety of Oslo stock exchange got affected at the same time as our chosen event and the NAS stock follows the same market principals.



Prior to the corona pandemic outbreak, NAS had already faced considerable financial distress (NAS 2019). The company is in heavily financial dept, and by 2019 they had a total dept of over 80 billion NOK (Rystad 2020). The company's debt-financed fleet and exposure to volatile fuel prices, coupled with intense price competition in the low-cost segment, resulted

in a fragile financial position even before the pandemic. In 2017-2019 the firm had a major deficit of just under 5 billion NOK (Rystad 2020). By looking at historical stock prices (Invest 2020), we can see a fire sale in 2019 from the firm's poor development, but this development met a stable price point around May 2019.

At the same time as analyzing the situation for NAS, we did an extended analysis for a cross-sectional sample with 9 more European companies. This was done to ensure the validity of t-testing, this topic is discussed later. Figure 3 shows all airline stocks used in the cross-sectional sample, with both OSEBX index and MSCI index. Here the price is changed from the original currencies to NOK for a better presentation, at the same time the y-axis is shown in log-format, this allows stocks at different price levels to be compared on the same plot. All prices in the cross-sectional sample were already reported in NOK in the LSEG export files, meaning no manual currency conversion was required. Thus, each price reflects the correct exchange rate of its respective trading day. The windows for the analysis stay the same.



As shown in figure 3, before the event, prices were broadly stable with a moderate development. During the event window we can observe a drop for all airline and index stock prices, which shows a similar development for the European market as the Norwegian market (NAS). After the event the recoveries are uneven, which can be explained by other factors that are not relevant for our analysis. However, NAS shows a steady decline after the event, which can indicate a late reaction to the event. Supporting this trend is the Iceland Air stock who also have a negative development after the event and similar market conditions.

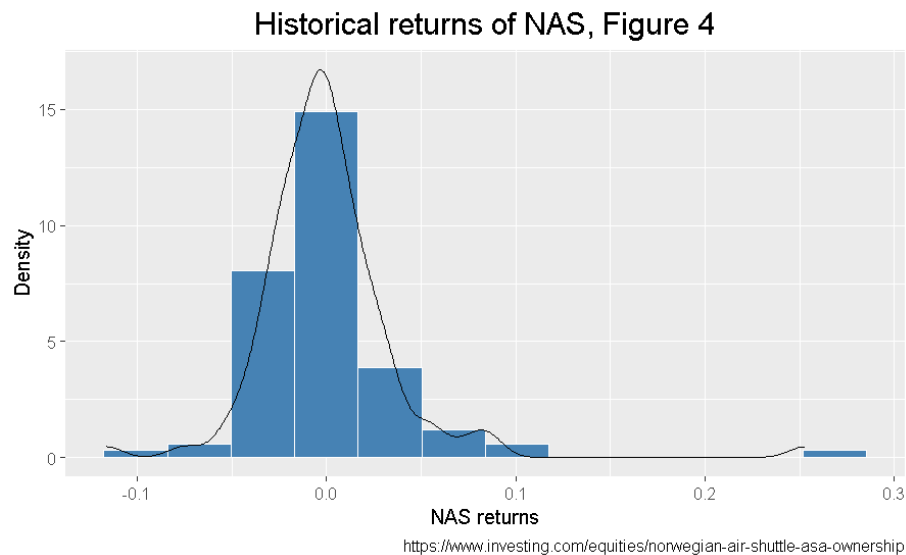


Figure 4 describes the empirical distribution of daily returns on NAS in the estimation window used to show the normal development in the stock, this is the normal return before the event, made with R-Studio. Returns are clustered around zero, indicating that NAS's returns exhibit small day to day changes with a rather normal distribution. We have a slight weighting left, meaning the normal return had a slight decline over time. At the same time, we can observe some outliers indicating some large return changes but only occurring occasionally.

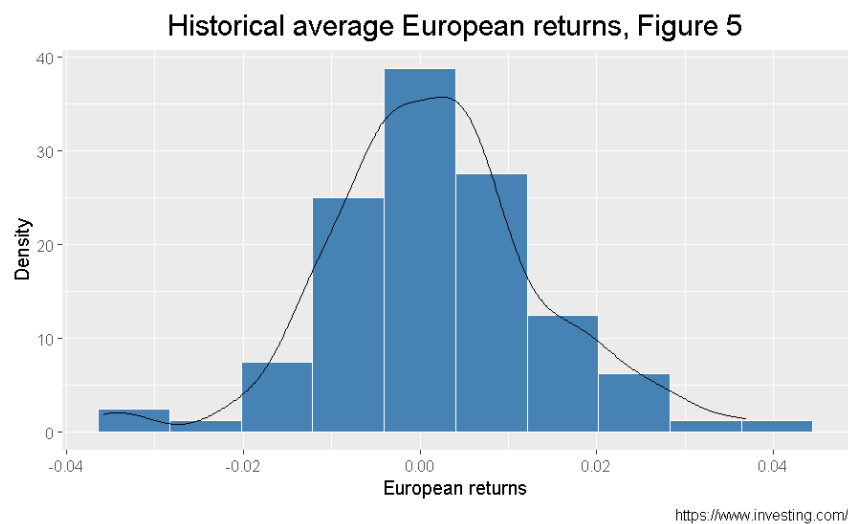


Figure 5 describes the empirical distribution of the multi-firm situation, where the data is an average return of the 9 European firms, made with R-Studio. Compared to the individual NAS return distribution, the tails here show fewer extreme observations, which is consistent with expectations of idiosyncratic risk when averaging several stock returns. Both graphs indicate a normal distribution of returns before the event happened.

Analysis & Discussion

Single company analysis

The CAPM model was estimated using daily data from the estimation window, which reflects normal market conditions and is isolated from the event itself. The regression of NAS excess returns on the OSEBX excess returns produced an alpha of -0.0005 and a beta of 1.78 . The near-zero alpha indicates no abnormal performance before the event, while the high beta confirms that the stock is highly sensitive to market fluctuations. The R^2 of 0.062 implies that most short-term variation is idiosyncratic, as expected for daily data. These estimates define the normal-return benchmark used to calculate expected and abnormal returns in the event window, see earlier chapters for reference.

OLS Regression results for Norwegian Air Shuttle

Variable	Results		
	Coefficient	Std.err	t-stat
Constant	-0.0005	0.004	-0.137
OSEBX excess	1.7795	0.482	3.695
R2	0.0620		
Observations	99.0000		

Note: Estimation window: 26 Sep 2019 - 19 Feb 2020

$$E[R_{it}|Event = 0] = -0,0005 + 1,7795[R_{mt} - R_{f,t}]$$

This model provides the benchmark return used to calculate and compare abnormal returns around the event. Applying the CAPM benchmark, daily abnormal returns were calculated using equation (8) and subsequently aggregated into cumulative abnormal returns using equation (10) across several event windows. The following results illustrate how NAS stock reacted to the WHO pandemic announcement on 11 March 2020.

Table 3 reports the cumulative abnormal returns for different event windows around March 11, 2020. Over the $[-10, +10]$ window, the CAR is -28.45% , indicating a sharp negative market reaction relative to the benchmark model. If markets were semi-strong efficient in the sense of Fama (1970), we would expect $CAR \approx 0$, implying no abnormal effect once information is public.

Table 3: Event Study Results for Norwegian Air Shuttle

Window	Results			
	n	Std. Err	CAR	t-stat
[-10, +10]	21	0.113	-28.45	-0.56
[-5, +5]	11	0.138	9.84	0.21
[-3, +3]	7	0.069	-24.01	-1.37
[-1, +1]	3	0.052	-14.16	-1.58

Note: CAR = Cumulative Abnormal Return. t-stat = t-statistic for event window significance.

In contrast, the [-5, +5] window shows a small positive CAR, which may reflect a short-term correction after the initial overreaction. The [-3, +3] window again yields a negative CAR similar in sign to the wider window, suggesting that the negative response was concentrated around the event. Finally, the immediate event window [-1, +1] shows a CAR of -14.16%, but the corresponding t-statistic is too low to reject the null hypothesis $H_0: CAR = 0$.

Following standard event-study conventions, we only reject H_0 when $|t| > 1.96$, meaning none of the windows reach statistical significance. Overall, while prices moved sharply around the event, the lack of statistical significance means we cannot conclude that the event had a systematic abnormal impact. This result is consistent with the semi-strong form of the Efficient Market Hypothesis, where prices rapidly incorporate public information.

Although the CAR values differ from zero across most event windows, the corresponding t-statistics indicate that these deviations are not statistically significant. This discrepancy can arise for several reasons. One likely explanation for the insignificant t-statistics is the relatively small sample size compared to what is typically used in event studies. In our case, the event windows contain between 3 and 21 trading-day observations, which is considerably smaller than what is typically used in event-study settings, where event windows often rely on 20–30 observations or large cross-sectional samples to ensure sufficient statistical power. A limited number of observations reduces the power of the t-test to detect statistical significance, even when the CAR and thus the economic effect appears large. The relatively large standard errors in Table 3 indicate high volatility, which directly lowers the t-statistics. The combination of unusually large standard errors, very wide fluctuations in CAR values, and the small number of trading day observations provides direct evidence of extreme market

uncertainty surrounding the event. Hence, the lack of statistical significance mainly reflects extreme market uncertainty rather than the absence of an economic effect.

Therefore, the lack of statistical significance does not necessarily imply that the event had no economic impact, but rather that the effect is difficult to isolate statistically given the extreme market conditions. The negative, insignificant CAR pattern suggests an initial overreaction to the event followed by rapid correction, consistent with a semi-strongly efficient market that quickly absorbs new information. To overcome these limitations, we included the analysis to a broader cross-section of European airlines ($N = 9$).

The need for robustness and validity

The extended analysis, incorporating nine additional European based firms, including some Nordic firms, aimed to overcome the limitations found in the NAS analysis. For each company, we calculate abnormal returns using the same procedure with some differences. Instead of using the OSEBX index, we use a broad MSCI Europe market index as the benchmark for expected returns. We then compute the Average Abnormal Return (AAR) for each day in the event window, which is the mean of all 10 companies' abnormal returns on that day. Followed by calculating the Cumulative Average Abnormal Return (CAAR) by averaging the CARs of all companies, as shown in earlier chapters.

This multi-firm approach provides a larger sample for statistical testing, which is effectively leveraging $N = 210$ observations instead of the earlier $N = 21$, where the latter gave weak statistical power. We performed a cross-sectional t-test on the CAAR to determine if, on average, the event had a significant impact on stock returns across the samples. By increasing the number of observations and using AAR/CAAR, the extended analysis improves the reliability of our inferences about the event's effect, which we can then use to examine NAS in a more robust environment.

Table 4: Event Study Results for ten European airlines

Window	n	Results	
		CAAR	t-stat
[-10, +10]	210	-14.97%	-3.330
[-5, +5]	110	-9.24%	-1.922
[-3, +3]	70	4.04%	0.940
[-1, +1]	30	3.94%	1.786

Note: CAAR = Cumulative average abnormal returns for different event windows around the WHO announcement on 11 March 2020

The results from Table 4 reveal a statistically significant negative CAAR of -14.97% for the $[-10, +10]$ window with $t = -3.23$, significant at the 1% level. This indicates that the decline in airline stocks was not limited to the immediate days around the announcement but persisted over a longer period, reflecting a broad reassessment of the aviation sector following the WHO declaration on the 13th of March 2020.

In contrast, the shorter windows $[-3, +3]$ and $[-1, +1]$ show smaller and statistically insignificant effects, suggesting that prices adjusted rapidly to the new information and stabilized thereafter. This pattern is consistent with the semi - strong form of market efficiency, where prices incorporate publicly available information without delay.

The extension from a single firm to a multi-firm event study was essential to improve robustness and validity. The single-company analysis of NAS, shown in Table 3, produced economically meaningful but statistically insignificant results due to firm-specific volatility and a limited number of observations. By including nine additional airlines, the analysis introduces cross-sectional variation that strengthens statistical power and filters out company-specific noise. The resulting CAAR therefore captures the systematic market component, providing a more credible and generalizable estimate of the event's impact.

It is worth noting that the sign of NAS's CAR differs slightly from the aggregate CAAR in some event windows, such as $[-5, +5]$. This difference reflects idiosyncratic behaviour rather than inconsistency. While NAS temporarily rebounded after the initial decline, the average effect across the industry remained negative, indicating that the market reacted negatively to the announcement. The multi-firm approach thus enhances both the robustness and external

validity of the findings, confirming that NAS reaction was part of a broader, systematic market adjustment.

Hypothesis evaluation & Conclusion

Having now established the empirical results for both the single-firm and multi-firm event studies, we can return to the main hypothesis of this study, whether the WHO announcement on March 11, 2020, had a value-relevant impact on NAS stock price.

The results in Table 4 show a statistically significant negative CAAR of -14.97% over the $[-10, +10]$ window ($t = -3.23$) indicating that airline stocks experienced a broad and persistent decline following the WHO announcement. For this wider event window, the null hypothesis of no abnormal performance for both CAR and CAAR can be rejected.

However, for the shorter windows $[-3, +3]$ and $[-1, +1]$, the t -statistics are insignificant, implying that the strongest market response occurred in the days immediately surrounding the event, followed by stabilization. This pattern suggests a rapid adjustment consistent with semi-strong market efficiency, but the extended negative reaction over the longer window points to a temporary overreaction or delayed reassessment of firm value.

Comparing the single-firm and multi-firm results underscores this point. The CAR for NAS alone was economically large but statistically weak, while the cross-sectional CAAR strengthened the signal and confirmed that the decline was industry-wide rather than firm-specific. The multi-firm approach thus improved robustness and validity by filtering out idiosyncratic noise and revealing the systematic component of the reaction.

The findings provide partial support for semi-strong efficiency. Prices reacted quickly to new public information, but the sustained negative CAAR over the longer horizon indicates that markets may have required additional time to fully process the implications of the pandemic for the airline industry. According to the efficient market hypothesis, stock prices should instantaneously reflect all available information, the strong negative CAAR indicates that the market responds negatively to the announcement. Earlier studies implies that air travel demand is highly income elastic (Martins & Cro 2022), which induces that economic negative development should be quickly reflected into reduced revenues for airline firms which was not the case in our result.

To explain why NAS stock appeared to react more slowly compared to the broader market and especially the international airline sample, three plausible factors can be identified. First, the most likely explanation is that the Norwegian government announced strict national lockdown measures on 12 March, one day after the WHO event. This subsequent announcement directly affected Norwegian's operations and may have overlapped with the global event, causing investors to wait for confirmation of how the pandemic would impact domestic travel before adjusting their valuations.

Secondly, Norwegian Air Shuttle's stock was characterized by low liquidity and high firm-specific risk. This meant that trading took longer, and prices adjusted less efficiently compared to larger, more liquid airline stocks. The substantial fall in stock prices reflects both the systematic and firm-specific risk. Other firm-specific characteristics like debt, size and business model are shown to have an impact on the reactions from the WHO announcement (Martin & Cro 2022), which can be contributing factors for the late reaction.

Thirdly the corona pandemic represented a macroeconomic shock that increased uncertainty (Altig et al 2020), the situation can be compared to a flight-to-quality period, where investors move away from risky assets as the airline stocks. Moreover, the longer response window for NAS can also be explained by the high levels of uncertainty, which made the investors choice of investment difficult.

Finally, the CAPM beta was estimated under calm pre-crisis conditions and may not fully capture the sudden surge in systematic risk during the event period. Consequently, part of the measured abnormal return might reflect model misspecification rather than delayed market response.

Taken together, the evidence is most consistent with the semi-strong form of market efficiency with frictions, indicating that public information was incorporated rapidly into prices. The multi-firm results show a broad, value-relevant reaction to the WHO announcement (significantly negative CAAR). However, the strongest declines appear over the wider $[-10, +10]$, $[-5, +5]$ windows, while shorter windows around $[-1, +1]$, $[-3, +3]$ are weak or even slightly positive. This pattern points to underreaction and post-event drift, implying that full repricing required several days. The single-firm results for NAS are noisier, mostly negative but less precisely estimated. Consistent with illiquidity, firm-specific uncertainty, overlapping domestic policy news, and limits to arbitrage that slow price

discovery. Thus, the findings go beyond weak-form efficiency but fall short of strong-form immediacy, best described as semi-strong efficiency with delayed information under high uncertainty. Because airline demand is highly income-elastic, the WHO announcement implied a disproportionate, income driven fall in future revenues which is consistent with semi-strong efficient market form, this was progressively incorporated into prices, producing a short window reaction but a large negative CAAR over the wider window.

Overall, the evidence points to semi-strong efficiency with frictions, where public news was incorporated, but the pandemic's cash-flow shock required several days to be fully priced, yielding a broad negative effect and a noisier and slower adjustment for NAS

References

Articles & documents:

- Fama, E. F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, 25(2), 383–417. <https://doi.org/10.2307/2325486>
- Fama, E. F & French, R. K (2004): The Capital Asset Pricing Model: Theory and Evidence, <https://www.aeaweb.org/articles?id=10.1257/0895330042162430>
- Fama, E. F & French, R. K (1996), Multifactor Explanations of Asset Pricing Anomalies, <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.1996.tb05202.x>
- Sharpe, F. W (1964), Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk. <https://www.jstor.org/stable/2977928>
- MacKinlay, C. A (1997), Event Studies in Economics and Finance, <https://www.jstor.org/stable/2729691?seq=1>
- Altig D et al (2020), Economic uncertainty before and during the COVID-19 pandemic, <https://www.sciencedirect.com/science/article/pii/S0047272720301389>
- Martins M. A. & Cro S (2022), Airline stock markets reaction to the COVID-19 outbreak and vaccines: An event study <https://pmc.ncbi.nlm.nih.gov/articles/PMC9393161/#sec2>
- Norwegian Air Shuttle ASA, (2019), Annual Report 2019, <https://www.norwegian.com/globalassets/ip/documents/about-us/company/investor-relations/reports-and-presentations/annual-reports/annual-report-norwegian-2019.pdf>

- Tretheway M.W & Markhvida K (2022), The aviation value chain: Economic returns and policy issues,
<https://www.sciencedirect.com/science/article/pii/S0969699714000854>
- Xue Yufei (2025), Lecture 2: Asset pricing and event study (I)
- Maneenop & Kotcharin (2020), The impacts of COVID-19 on the global airline industry: An event study approach,
<https://www.sciencedirect.com/science/article/pii/S0969699720305032?via%3Dihub>
- World Health Organization (2020), Coronavirus disease (COVID-19) pandemic, declaration, <https://www.who.int/europe/emergencies/situations/covid-19>
- ChatGPT, AI Generator, <https://chatgpt.com/>, used for translating text from Norwegian to English.

Data structures:

- Statistisk Sentralbyrå, «08503: Lufttransport. Flygingar og seter, etter lufthamn, trafikktype og innanlands-/utanlandsflyging 2009M01 - 2025M08», Lyle & Pilskog, <https://www.ssb.no/statbank/table/08503>, last retrieved: 03.10.2025
- Investing.com, <https://www.investing.com/>, last retrieved: 03.10.2025. «Norwegian Air Shuttle Stock Price History», «SAS Stock Stock Price History», «Aegean Airlines Stock Price History», «Air France KLM Stock Price History», «EasyJet Stock Price History», «Wizz Air Holdings PLC (WIZZ)», «Deutsche Lufthansa AG (LHAG) », «Ryanair Holdings PLC ADR (RYAAY)», «Tui AG NA (TUI1n)», «Icelandair Group hf. (ICEAIR)», «MSCI World (MIWO000000PUS)», «OSE Benchmark (OSEBX)»,
- Yahoo Finance, «Norwegian Air Shuttle ASA (NAS.OL)», last retrieved: 02.10.2025, <https://finance.yahoo.com/quote/NAS.OL/>
- Refinitiv, <https://www.lseg.com/>, last retrieved; 07.10.2025. «Norwegian Air Shuttle ASA», «NOWA=NRGB», «AIR FRANCE KLM SA», «MSCI Europe Price Index USD end of day», «Aegean Airlines SA», «Icelandair Group hf», «Easyjet PLC», «Deutsche Lufthansa AG», «Ryanair Holdings PLC», «SAS AB», «TUI AG», «Wizz Air Holdings PLC», «Oslo Exchange Benchmark Index_GI».

Code/Extras

Python Code for calculating CAPM:

```
"""
```

CAPM estimate for Norwegian Air Shuttle versus the OSEBX market index.

```

# Notes for instructors:
# - Reads Reuters/Refinitiv Excel exports (same layout as the other airline files).
# - Computes daily returns, excess returns, and runs an OLS regression on the estimation
  window.
# - Reports Jensen's alpha and beta alongside the usual regression summary.
"""

from __future__ import annotations

from pathlib import Path

import numpy as np
import pandas as pd
import statsmodels.api as sm

# --- Configuration -----

DATA_DIR = Path("Flyselskap_data")
NAS_FILE = "nas.xlsx"
OSEBX_FILE = "oslo_bors.xlsx"
RF_FILE = "nowa_rf.xlsx"
DAY_COUNT = 252 # switch to 360 if you prefer the money-market convention
SAMPLE_START = "2019-09-26"
SAMPLE_END = "2020-02-19"

# --- Helper functions -----

def read_price_history(path: Path) -> pd.Series:
    """Return the closing-price series from a Reuters export."""
    preview = pd.read_excel(path, header=None, nrows=80)
    header_row = int(
        preview.index[preview.iloc[:, 0].astype(str).str.contains("Exchange Date")][0]

```

```

)
df = pd.read_excel(path, header=header_row)[["Exchange Date", "Close"]].dropna()
df["Exchange Date"] = pd.to_datetime(df["Exchange Date"])
df["Close"] = pd.to_numeric(df["Close"], errors="coerce")
return df.dropna().set_index("Exchange Date").sort_index()["Close"]

def load_risk_free(path: Path, day_count: int) -> pd.Series:
    """Load NOWA and convert the percentage quote into a daily decimal rate."""
    rf = pd.read_excel(path)
    rf = rf.rename(columns={rf.columns[0]: "Date", rf.columns[1]: "rate"})
    rf["Date"] = pd.to_datetime(rf["Date"])
    rf["rate"] = pd.to_numeric(rf["rate"], errors="coerce")
    rf = rf.dropna(subset=["rate"]).set_index("Date").sort_index()
    rf = rf[(rf.index >= SAMPLE_START) & (rf.index <= SAMPLE_END)]
    rf["rf_daily"] = rf["rate"] / 100 / day_count
    return rf["rf_daily"]

# --- Analysis -----

nas_prices = read_price_history(DATA_DIR / NAS_FILE)
osebx_prices = read_price_history(DATA_DIR / OSEBX_FILE)
rf_daily = load_risk_free(DATA_DIR / RF_FILE, day_count=DAY_COUNT)

returns = pd.concat(
    [
        nas_prices.pct_change().rename("norwegian_return"),
        osebx_prices.pct_change().rename("osebx_return"),
        rf_daily.rename("rf"),
    ],
    axis=1,
).dropna()

returns = returns.loc[SAMPLE_START:SAMPLE_END]

```

```

returns["norwegian_excess"] = returns["norwegian_return"] - returns["rf"]
returns["osebx_excess"] = returns["osebx_return"] - returns["rf"]

X = sm.add_constant(returns["osebx_excess"])
y = returns["norwegian_excess"]
capm_model = sm.OLS(y, X).fit()

alpha = capm_model.params["const"]
beta = capm_model.params["osebx_excess"]

print(f'Observations used in the regression: {len(returns):d}')
print(f'Alpha (daily Jensen's alpha): {alpha:.6f}')
print(f'Beta (CAPM beta): {beta:.4f}')
print("\n--- OLS summary ---")
print(capm_model.summary())

```

Code for calculating CAR:

```

# --- Import libraries for data handling, date offsets, file paths, statistics ---
import pandas as pd
from pandas.tseries.offsets import BDay # business-day window selection
from pathlib import Path
from math import sqrt
from scipy import stats # two-sided t distribution for significance tests

# --- Model calibration and event definition ---
alpha = -0.0005 # intercept from the CAPM previously estimated
beta = 1.7795 # previously estimated market beta estimated over the estimation window
event_date = pd.Timestamp("2020-03-11") # WHO pandemic announcement (Norwegian Air
focus date)

# --- Data ingestion: Norwegian Air, OSEBX, and risk-free rate ---
data_dir = Path("Data")

```

```

# Norwegian Air share prices from Yahoo Finance export; rename 'Price' column to Date
nas = (
    pd.read_csv(data_dir / "norwegian.csv", skiprows=[1, 2])
    .rename(columns={"Price": "Date"})
    .assign(
        Date=lambda df: pd.to_datetime(df["Date"]),
        Close=lambda df: pd.to_numeric(df["Close"], errors="coerce"),
    )
    .dropna(subset=["Close"])
    .set_index("Date")
    .sort_index()
)

# Oslo Benchmark Index (OSEBX); same cleaning steps for consistency
osebx = (
    pd.read_csv(data_dir / "osebx_index.csv", skiprows=[1, 2])
    .rename(columns={"Price": "Date"})
    .assign(
        Date=lambda df: pd.to_datetime(df["Date"]),
        Close=lambda df: pd.to_numeric(df["Close"], errors="coerce"),
    )
    .dropna(subset=["Close"])
    .set_index("Date")
    .sort_index()
)

# Daily Norwegian risk-free rate (already in decimal form, e.g. 0.0003 = 0.03%)
rf = (
    pd.read_csv(data_dir / "Norway_Rf_daily.csv", skiprows=1)
    .rename(columns={"date": "Date", "Rf(1d)": "rf"})
    .assign(
        Date=lambda df: pd.to_datetime(df["Date"], format="%Y%m%d"),
        rf=lambda df: pd.to_numeric(df["rf"], errors="coerce"),
    )
)

```

```

)
.dropna(subset=["rf"])
.set_index("Date")
.sort_index()
)

# --- Construct daily returns aligned on dates ---
returns = (
    pd.concat(
        [
            nas["Close"].pct_change().rename("norwegian_return"),
            osebx["Close"].pct_change().rename("osebx_return"),
            rf["rf"],
        ],
        axis=1,).dropna())

# --- Event windows, abnormal returns, and significance testing ---
summary = []
for k in [10, 5, 3, 1]: # evaluate several symmetric event windows
    win = returns.loc[event_date - BDay(k): event_date + BDay(k)].copy()

    # Excess returns over the risk-free rate
    win["norwegian_excess"] = win["norwegian_return"] - win["rf"]
    win["osebx_excess"] = win["osebx_return"] - win["rf"]

    # Expected return from CAPM and abnormal return (AR)
    win["expected_capm"] = alpha + beta * win["osebx_excess"]
    ar = win["norwegian_excess"] - win["expected_capm"]

    # Descriptive statistics and t-test of mean AR
    n = len(ar)
    mean_ar = ar.mean()
    std_ar = ar.std(ddof=1)
    t_stat = mean_ar / (std_ar / sqrt(n)) if std_ar > 0 else float("nan")

```

```

p_val = 2 * stats.t.sf(abs(t_stat), df=n - 1) if std_ar > 0 else float("nan")

summary.append(
    {
        "window": f"[-{k}, +{k}]",
        "n": n,
        "mean_AR": mean_ar,
        "std_AR": std_ar,
        "CAR": ar.sum(),
        "t_stat": t_stat,
        "p_value": p_val,
    }
)

# --- Output results ---
results = pd.DataFrame(summary)
print(results)

```

Code for calculating CAAR:

```

"""
Daily event study for European airlines measured in NOK.

```

Method:

- 1) Estimate a simple market model (CAPM) in [-105, -5] business days pre-event.
- 2) Compute abnormal returns (AR).
- 3) Aggregate to AAR/CAAR over firms.
- 4) Test CAAR across firms with a t-test.

Notes:

- All returns are in NOK.
- Risk-free rate is NOK (NOWA daily fixing), converted to a daily rate using 252.

```

"""

```

```

from pathlib import Path
from typing import Dict, List, Tuple

import numpy as np
import pandas as pd
import statsmodels.api as sm
from scipy import stats

# ----- Config ----- #

DATA_DIR = Path("Flyselskap_data")

AIRLINE_FILES: List[str] = [
    "aeg_greece.xlsx",
    "air_france.xlsx",
    "easyjet_uk.xlsx",
    "iceland_air.xlsx",
    "lufthansa.xlsx",
    "nas.xlsx",
    "ryanair.xlsx",
    "sas.xlsx",
    "tui.xlsx",
    "wizz_air_uk.xlsx",
]

MARKET_FILE = "europa_index.xlsx" # Europe index expressed in NOK (prefer TR)
RF_FILE = "nowa_rf.xlsx"          # NOWA daily fixing: (Date, Value) in % p.a.

EVENT_DATE = pd.Timestamp("2020-03-11") # WHO announcement (t = 0)

# Estimation window: [-105, -5] business days before the event (avoid leakage)
ESTIMATION_OFFSETS: Tuple[int, int] = (-105, -5)

# Event windows to report (relative to t=0)

```



```

EVENT_WINDOWS: Dict[str, Tuple[int, int]] = {
    "±10": (-10, 10),
    "±5": (-5, 5),
    "±3": (-3, 3),
    "±1": (-1, 1),
}

DAY_COUNT = 252 # Daily conversion from annualized rates (market convention)

# ----- IO Utilities ----- #

def read_exchange_history(path: Path) -> pd.Series:
    """
    Read a LSEG/Reuters-style Excel with 'Exchange Date' and 'Close' columns.

    The header row may not be the first row (typical RIC exports).
    Returns
    -----
    pd.Series
        Close prices, indexed by pandas Timestamp, sorted ascending.
    """
    preview = pd.read_excel(path, header=None, nrows=80)
    header_row = int(
        preview.index[
            preview.iloc[:, 0].astype(str).str.contains("Exchange Date")
        ][0]
    )
    df = pd.read_excel(path, header=header_row)[["Exchange Date", "Close"]].dropna()
    df["Exchange Date"] = pd.to_datetime(df["Exchange Date"])
    df["Close"] = pd.to_numeric(df["Close"], errors="coerce")
    return (
        df.dropna()
        .set_index("Exchange Date")
    )

```

```

        .sort_index()["Close"]
    )

def load_risk_free(path: Path, day_count: int = DAY_COUNT) -> pd.Series:
    """
    Load NOK risk-free series (NOWA daily fixing) and convert % p.a. to daily.

    Expected Excel layout: first column = Date, second = Value (% p.a.).
    Returns
    -----
    pd.Series
        Daily risk-free rate aligned by date (no calendar mapping yet).
    """
    rf = pd.read_excel(path)
    rf = rf.rename(columns={rf.columns[0]: "date", rf.columns[1]: "rate"})
    rf = rf.dropna(subset=["date", "rate"])
    rf["date"] = pd.to_datetime(rf["date"])
    rf["rate"] = pd.to_numeric(rf["rate"], errors="coerce")
    rf = rf.dropna().set_index("date").sort_index()
    # Simple daily rate; for short windows this is equivalent to compounding
    rf["rf_daily"] = rf["rate"] / 100 / day_count
    return rf["rf_daily"]

# ----- Core Methods ----- #

def compute_capm_abnormal(
    close: pd.Series,
    market_returns: pd.Series,
    rf_series: pd.Series,
    event_date: pd.Timestamp,
    est_offsets: Tuple[int, int],
) -> pd.DataFrame:
    """

```

Estimate CAPM in the estimation window and compute abnormal returns (AR).

Parameters

close : pd.Series

Stock close prices (NOK).

market_returns : pd.Series

Market returns (NOK).

rf_series : pd.Series

Daily NOK risk-free rate.

event_date : pd.Timestamp

Event date ($t = 0$).

est_offsets : (int, int)

Business-day offsets relative to event_date, e.g. (-105, -5).

Returns

pd.DataFrame

DataFrame with columns: stock_return, market_return, rf, excess_stock, excess_market, abnormal. Index is trading dates.

"""

```
stock_ret = close.pct_change().dropna().rename("stock_return")
```

```
aligned = (
```

```
    pd.concat(
```

```
        [stock_ret, market_returns.rename("market_return"),
```

```
        rf_series.rename("rf")],
```

```
        axis=1,
```

```
        join="inner",
```

```
    )
```

```
    .dropna()
```

```
)
```

```
aligned["excess_stock"] = aligned["stock_return"] - aligned["rf"]
```

```
aligned["excess_market"] = aligned["market_return"] - aligned["rf"]
```

```

# Estimation window by calendar (works if calendars are aligned).
est_start = event_date + pd.offsets.BDay(est_offsets[0])
est_end = event_date + pd.offsets.BDay(est_offsets[1])
estimation = aligned.loc[est_start:est_end]

X = sm.add_constant(estimation["excess_market"])
model = sm.OLS(estimation["excess_stock"], X).fit()
alpha = float(model.params["const"])
beta = float(model.params["excess_market"])

aligned["abnormal"] = (
    aligned["excess_stock"] - (alpha + beta * aligned["excess_market"])
)
return aligned

def extract_window(
    series: pd.Series,
    event_date: pd.Timestamp,
    start: int,
    end: int,
) -> pd.Series:
    """
    Extract an event window using business-day calendar around event_date.

    Returns a series indexed by event time  $\tau \in [\text{start}, \text{end}]$  with  $\tau=0$  at event.
    """
    idx = pd.bdate_range(
        event_date + pd.offsets.BDay(start),
        event_date + pd.offsets.BDay(end),
    )
    window = series.reindex(idx)
    window.index = range(start, end + 1)

```

```

return window

# ----- Pipeline ----- #

def main() -> None:
    # Market: price series → daily returns (NOK)
    market_prices = read_exchange_history(DATA_DIR / MARKET_FILE)
    market_returns = market_prices.pct_change().dropna()

    # Risk-free: NOWA daily → align to market calendar to avoid uneven joins
    rf_daily = load_risk_free(DATA_DIR / RF_FILE, day_count=DAY_COUNT)
    rf_daily = rf_daily.reindex(market_returns.index).ffill()

    window_panels: Dict[str, List[pd.Series]] = {k: [] for k in EVENT_WINDOWS}
    aar_curves: Dict[str, pd.Series] = {}
    caar_curves: Dict[str, pd.Series] = {}
    skipped: Dict[str, str] = {}

    # Firm-level AR computation and window extraction
    for fname in AIRLINE_FILES:
        try:
            close = read_exchange_history(DATA_DIR / fname)
            abnormal_df = compute_capm_abnormal(
                close,
                market_returns,
                rf_daily,
                EVENT_DATE,
                ESTIMATION_OFFSETS,
            )
            firm = Path(fname).stem
            for label, (start, end) in EVENT_WINDOWS.items():
                window_panels[label].append(
                    extract_window(

```

```

        abnormal_df["abnormal"], EVENT_DATE, start, end
    ).rename(firm)
)
except Exception as exc: # keep going and report which series failed
    skipped[fname] = str(exc)

# Cross-sectional AAR/CAAR and t-tests on CAR
summary_rows: List[Dict[str, float]] = []
for label, series_list in window_panels.items():
    window_df = pd.concat(series_list, axis=1)

    # Average across firms each event day (AAR), then cumulative (CAAR curve)
    aar = window_df.mean(axis=1)
    aar_curves[label] = aar
    caar_curves[label] = aar.cumsum()

    # Firm CARs over the window → CAAR (mean of CARs) and t-test
    car = window_df.sum(axis=0) # CAR per firm
    n = car.count()
    caar_mean = car.mean()
    std = car.std(ddof=1)
    t_stat = np.nan if n < 2 or std == 0 else caar_mean / (std / np.sqrt(n))
    p_val = np.nan if n < 2 or std == 0 else stats.t.sf(abs(t_stat), df=n - 1) * 2

    summary_rows.append(
        {
            "window": label,
            "firms": int(n),
            "CAAR": float(caar_mean),
            "t": float(t_stat) if pd.notna(t_stat) else np.nan,
            "p": float(p_val) if pd.notna(p_val) else np.nan,
            "event_AAR": float(aar.loc[0]),
        }
    )

```

```

# store concatenated frame for later inspection/printing
window_panels[label] = window_df

summary = pd.DataFrame(summary_rows).set_index("window").sort_index()

if skipped:
    print("Skipped series:", skipped)

# Pretty print (percent for CAAR and event-day AAR)
summary_view = summary.copy()
summary_view[["CAAR", "event_AAR"]] *= 100
summary_view = summary_view.round({"CAAR": 2, "event_AAR": 2, "t": 3, "p": 3})

print("CAAR t-tests (values in % where noted):")
print(summary_view)

print("\nEvent-day abnormal returns (%):")
event_day = window_panels["±10"].loc[[0]].T.rename(columns={0: "AR_t0"})
event_day["AR_t0"] = (event_day["AR_t0"] * 100).round(2)
print(event_day.sort_values("AR_t0"))

if __name__ == "__main__":
    main()

```

RStudio graphs and tables:

```

# picking up relevant packages
library(tidyverse)
library(ggplot2)
library(readr)

```

```

library(readxl)
library(janitor)
library(lubridate)
library(scales)
library(dplyr)
library(knitr)

# saving the dataset of interest, link: https://www.ssb.no/statbank/table/08503
flights <- read_xlsx("C:\\Users\\Eier\\Downloads\\08503_20251002-171638.xlsx")

flights <- flights |>
  select(-starts_with("0"), -"...2", -"...3", -"...4") |> # removing unnecessary column
  drop_na() |> # dropping all NA's
  rename_at(1, ~"month") |> # changing column name with column index
  rename_at(2, ~"flights") |>
  rename_at(3, ~"seats") |>
  mutate(flights = as.numeric(flights),
         seats = as.numeric(seats),
         month = ym(gsub("M", "-", month))) # changing row values and column class

# giving class and setup to month column
flights$month <- as.Date(paste0(substr(flights$month, 1, 4), "-",
                                     substr(flights$month, 6, 7), "-01"))

ggplot(flights, aes(x = month, y = seats)) +
  geom_bar(stat = "identity", width = 32, fill = "steelblue") + # plotting
  geom_vline(xintercept = as.Date("2020-03-11"), color = "red", size=0.7) + # indicator
  geom_label(x = as.Date("2020-03-11"), y = 4500000, label = "2020/03/11",
            fill = "red", size = 3.5) + # indicator label/number
  labs(title = "Total Commercial Airline Seats, Figure 1",
       subtitle = "Domestic and international flights",
       x = "Year",
       y = "Number of seats",
       caption = "https://www.ssb.no/statbank/table/08503") + # titles

```



```

scale_y_continuous(labels = label_number()) + # choosing form for number
theme(axis.text.x=element_text(angle = 0),
      legend.title=element_blank(),
      plot.title = element_text(hjust = 0.5, size = 17)) + # titles
scale_x_date(date_breaks = "12 month", date_labels = "%Y") # tick marks

# saving the dataset of interest, link: https://www.investing.com/indices/ose-benchmark-historical-data
osebx <- read.csv("C:\\Users\\Eier\\Downloads\\OSE Benchmark Historical Data.csv")

osebx <- osebx %>% # renaming variable names
  rename(date = 'X...Date',
         price_o = Price,
         open_o = Open,
         high_o = High,
         low_o = Low,
         vol_o = 'Vol.',
         changeo = 'Change..') %>% # changing observations with commas
  mutate(across(c(price_o, open_o, high_o, low_o),
                ~ as.numeric(gsub(",", "", .))))

# changing class to date
osebx$date <- mdy(osebx$date)

# saving the dataset of interest, link: https://www.investing.com/equities/norwegian-air-shuttle-asa-ownership
nas19 <- read.csv("C:\\Users\\Eier\\Downloads\\Norwegian Air Shuttle Stock Price History
(3).csv")

nas19 <- nas19 %>% # renaming variable names
  rename(date = 'X...Date',
         price = Price,
         open = Open,

```

```

    high = High,
    low = Low,
    vol = 'Vol.',
    change = 'Change..') %>% # changing observations with commas
mutate(across(c(price, open, high, low),
  ~ as.numeric(gsub(",", "", .))))

# changing class to date
nas19$date <- mdy(nas19$date)

# combining with OSEBX index
merge_nas_osebx <- left_join(nas19, osebx, by = "date")

ggplot(merge_nas_osebx, aes(x = date, y = price)) + # plotting
  geom_line(size = 0.8, color = "steelblue") +
  geom_line(aes(x = date, y = price_o), size = 0.8, color = "gold") +
  #geom_area(fill = "steelblue") + # filling area
  geom_vline(xintercept = as.Date("2019-09-26"), color = "green4", size = 0.7) + #
indicators
  geom_vline(xintercept = as.Date("2020-02-19"), color = "green4", size = 0.7) +
  geom_vline(xintercept = as.Date("2020-02-26"), color = "red4", size = 0.7) +
  geom_vline(xintercept = as.Date("2020-03-25"), color = "red4", size = 0.7) +
  geom_ribbon(aes(xmin = as.Date("2019-09-26"), xmax = as.Date("2020-02-19")), fill =
"green", alpha = 0.3) + # shading
  geom_ribbon(aes(xmin = as.Date("2020-02-26"), xmax = as.Date("2020-03-25")), fill =
"red", alpha = 0.3)) + # text
  annotate("text", x = as.Date("2019-09-15"), y = 4000, label = "Estimation window", color =
"black", size = 3, angle = 90) +
  annotate("text", x = as.Date("2020-04-02"), y = 4000, label = "Event window", color =
"black", size = 3, angle = 90) +
  annotate("text", x = as.Date("2019-04-01"), y = 4000, label = "NAS price", color =
"steelblue", size = 3) +
  annotate("text", x = as.Date("2019-04-01"), y = 1200, label = "OSEBX index", color =
"gold", size = 3) +

```

```

labs(title = "Historical NAS Prices & OSEBX Index, Figure 2",
      x = "Date",
      y = "Stock Price",
      caption = "https://www.investing.com/equities/norwegian-air-shuttle-asa-ownership") + # titles
scale_y_continuous(labels = label_number(), expand = c(0, 0)) + # choosing form for number
scale_x_continuous(expand = c(0, 0)) +
theme(axis.text.x=element_text(angle = 0),
      legend.title=element_blank(),
      legend.position = "none",
      plot.title = element_text(hjust = 0.5, size = 17)) + # titles
scale_x_date(date_breaks = "3 month") # tick marks

# selecting periods
nas1920 <- merge_nas_osebx %>%
  filter(date < "2020-02-26") %>%
  mutate(Period = "Pre-event") # adding label

nas2021 <- merge_nas_osebx %>%
  filter(date > "2020-03-25") %>%
  mutate(Period = "Post-event") # adding label

nasevent <- merge_nas_osebx %>%
  filter(date >= "2020-02-26" & date <= "2020-03-25") %>%
  mutate(Period = "Event window") # adding label

# we have a gap window with about 5 days

nasestimatew <- merge_nas_osebx %>%
  filter(date >= "2019-09-26" & date <= "2020-02-19") %>%
  mutate(Period = "Estimation window") # adding label

```

```

nassum <- bind_rows(nas1920, nasevent, nas2021, nasestimatew) # combining

# summarize table
summary_table <- nassum %>%
  group_by(Period) %>%
  summarise(
    Mean = mean(price, na.rm = TRUE),
    Median = median(price, na.rm = TRUE),
    Min = min(price, na.rm = TRUE),
    Max = max(price, na.rm = TRUE),
    Days = n()
  )

summary_table2 <- nassum %>%
  group_by(Period) %>%
  summarise(
    Mean = mean(price_o, na.rm = TRUE),
    Median = median(price_o, na.rm = TRUE),
    Min = min(price_o, na.rm = TRUE),
    Max = max(price_o, na.rm = TRUE),
    Days = n()
  )

# extra package
library(kableExtra)

# table design
table1 <- kable(summary_table, format = "html", caption = "Summary for NAS") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = F) %>%
  kable_classic() %>%
  row_spec(0, bold = T) %>%
  footnote(general = "Data: https://www.investing.com/equities/norwegian-air-shuttle-asa-ownership ")

```

```

table2 <- kable(summary_table2, format = "html", caption = "Summary for OSEBX") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = F) %>%
  kable_classic() %>%
  row_spec(0, bold = T) %>%
  footnote(general = "Data: https://www.investing.com/indices/ose-benchamrk-historical-
data")

kables(list(table1, table2), format = "html")

table1
table2

# making returns with price t and price t-1
nasreturn <- nas19 %>%
  arrange(desc(date)) %>%
  mutate(ret = (price - lead(price)) / lead(price)) %>%
  filter(date >= as.Date("2019-09-26"),
         date <= as.Date("2020-02-19"))

# 19 februar 2020, 218, 26 september 2019 317

# plotting
ggplot(nasreturn, aes(x = ret)) + # histogram
  geom_histogram(aes(y = after_stat(density)),
                bins = 12, fill = "steelblue", color = "white") + # density line
  geom_density(linewidth = 0.5) + # titles
  labs(title = "Historical returns of NAS, Figure 4",
       x = "NAS returns",
       y = "Density",
       caption = "https://www.investing.com/equities/norwegian-air-shuttle-asa-ownership") + #
titles
  theme(axis.text.x=element_text(angle = 0),
        legend.title=element_blank(),
        legend.position = "none",

```

```

plot.title = element_text(hjust = 0.5, size = 17))

# saving the datasets of interest

# Aegean, link: https://www.investing.com/equities/aegean-airlines
aegean <- read.csv("C:\\Users\\Eier\\Downloads\\Aegean Airlines Stock Price History.csv")

aegean <- aegean %>% # renaming variable names
  rename(date = 'X...Date',
    price_aegean = Price,
    open_aegean = Open,
    high_aegean = High,
    low_aegean = Low,
    vol_aegean = 'Vol.',
    change_aegean = 'Change..') %>% # changing observations with commas
  mutate(across(c(price_aegean, open_aegean, high_aegean, low_aegean),
    ~ as.numeric(gsub(",", "", .)))) %>%
  mutate(price_aegean = price_aegean * 11.6) # changing to NOK

# changing class to date
aegean$date <- mdy(aegean$date)

# France, link: https://www.investing.com/equities/air-france---klm-historical-data
france <- read.csv("C:\\Users\\Eier\\Downloads\\Air France KLM Stock Price History.csv")

france <- france %>% # renaming variable names
  rename(date = 'X...Date',
    price_france = Price,
    open_france = Open,
    high_france = High,
    low_france = Low,
    vol_france = 'Vol.',
    change_france = 'Change..') %>% # changing observations with commas
  mutate(across(c(price_france, open_france, high_france, low_france),

```

```

    ~ as.numeric(gsub(",", "", .)))) %>%
mutate(price_france = price_france * 11.6) # changing to NOK

# changing class to date
france$date <- mdy(france$date)

# Easyjet, link: https://www.investing.com/equities/easyjet-historical-data
easyjet <- read.csv("C:\\Users\\Eier\\Downloads\\EasyJet Stock Price History.csv")

easyjet <- easyjet %>% # renaming variable names
  rename(date = 'X...Date',
    price_easyjet = Price,
    open_easyjet = Open,
    high_easyjet = High,
    low_easyjet = Low,
    vol_easyjet = 'Vol.',
    change_easyjet = 'Change..') %>% # changing observations with commas
mutate(across(c(price_easyjet, open_easyjet, high_easyjet, low_easyjet),
  ~ as.numeric(gsub(",", "", .)))) %>%
mutate(price_easyjet = price_easyjet * 13.4) # changing to NOK

# changing class to date
easyjet$date <- mdy(easyjet$date)

# Wizz, link: https://www.investing.com/equities/wizz-air-holdings-plc
wizz <- read.csv("C:\\Users\\Eier\\Downloads\\Wizz Air Holdings Stock Price History.csv")

wizz <- wizz %>% # renaming variable names
  rename(date = 'X...Date',
    price_wizz = Price,
    open_wizz = Open,
    high_wizz = High,
    low_wizz = Low,
    vol_wizz = 'Vol.',

```

```

      change_wizz = 'Change..') %>% # changing observations with commas
mutate(across(c(price_wizz, open_wizz, high_wizz, low_wizz),
      ~ as.numeric(gsub(",", "", .)))) %>%
mutate(price_wizz = price_wizz * 13.4) # changing to NOK

# changing class to date
wizz$date <- mdy(wizz$date)

# Lufthansa, link: https://www.investing.com/equities/dt-lufthansa
lufthansa <- read.csv("C:\\Users\\Eier\\Downloads\\Lufthansa Stock Price History.csv")

lufthansa <- lufthansa %>% # renaming variable names
  rename(date = 'X...Date',
    price_lufthansa = Price,
    open_lufthansa = Open,
    high_lufthansa = High,
    low_lufthansa = Low,
    vol_lufthansa = 'Vol.',
    change_lufthansa = 'Change..') %>% # changing observations with commas
mutate(across(c(price_lufthansa, open_lufthansa, high_lufthansa, low_lufthansa),
      ~ as.numeric(gsub(",", "", .)))) %>%
mutate(price_lufthansa = price_lufthansa * 11.6) # changing to NOK

# changing class to date
lufthansa$date <- mdy(lufthansa$date)

# Ryanair, link: https://www.investing.com/equities/ryanair-hldgs
ryanair <- read.csv("C:\\Users\\Eier\\Downloads\\Ryanair ADR Stock Price History.csv")

ryanair <- ryanair %>% # renaming variable names
  rename(date = 'X...Date',
    price_ryanair = Price,
    open_ryanair = Open,
    high_ryanair = High,

```



```

    low_ryanair = Low,
    vol_ryanair = 'Vol.',
    change_ryanair = 'Change..') %>% # changing observations with commas
mutate(across(c(price_ryanair, open_ryanair, high_ryanair, low_ryanair),
  ~ as.numeric(gsub(",", "", .)))) %>%
mutate(price_ryanair = price_ryanair * 10) # changing to NOK

# changing class to date
ryanair$date <- mdy(ryanair$date)

# TUI, link: https://www.investing.com/equities/tui-n
tui <- read.csv("C:\\Users\\Eier\\Downloads\\Tui Stock Price History.csv")

tui <- tui %>% # renaming variable names
  rename(date = 'X...Date',
    price_tui = Price,
    open_tui = Open,
    high_tui = High,
    low_tui = Low,
    vol_tui = 'Vol.',
    change_tui = 'Change..') %>% # changing observations with commas
mutate(across(c(price_tui, open_tui, high_tui, low_tui),
  ~ as.numeric(gsub(",", "", .)))) %>%
mutate(price_tui = price_tui * 11.6) # changing to NOK

# changing class to date
tui$date <- mdy(tui$date)

# Iceland, link: https://www.investing.com/equities/icelandair-group
iceland <- read.csv("C:\\Users\\Eier\\Downloads\\Icelandair Group Stock Price History.csv")

iceland <- iceland %>% # renaming variable names
  rename(date = 'X...Date',
    price_iceland = Price,

```

```

    open_iceland = Open,
    high_iceland = High,
    low_iceland = Low,
    vol_iceland = 'Vol.',
    change_iceland = 'Change..') %>% # changing observations with commas
mutate(across(c(price_iceland, open_iceland, high_iceland, low_iceland),
  ~ as.numeric(gsub(",", "", .))))

# changing class to date
iceland$date <- mdy(iceland$date)

# MSCI index, link: https://www.investing.com/indices/msci-world-historical-data
msci <- read.csv("C:\\Users\\Eier\\Downloads\\MSCI World Historical Data.csv")

msci <- msci %>% # renaming variable names
  rename(date = 'X...Date',
    price_msci = Price,
    open_msci = Open,
    high_msci = High,
    low_msci = Low,
    vol_msci = 'Vol.',
    change_msci = 'Change..') %>% # changing observations with commas
mutate(across(c(price_msci, open_msci, high_msci, low_msci),
  ~ as.numeric(gsub(",", "", .)))) %>%
mutate(price_msci = price_msci * 10) # changing to NOK

# changing class to date
msci$date <- mdy(msci$date)

# combining
dfs <- list(aegean, france, easyjet, wizz, lufthansa, ryanair, tui, iceland, msci, sas19, nas19,
  osebx)

# saving new dataset

```

```

europe <- reduce(dfs, left_join, by = "date")

ggplot(europe, aes(x = date, y = price)) + # plotting
  annotate(
    "rect",
    xmin = as.Date("2020-02-26"),
    xmax = as.Date("2020-03-25"),
    ymin = 0.9, ymax = Inf,    # covering y-axis, shading
    fill = "red", alpha = 0.3
  ) +
  annotate(
    "rect",
    xmin = as.Date("2019-09-26"),
    xmax = as.Date("2020-02-19"),
    ymin = 0.9, ymax = Inf,    # covering y-axis, shading
    fill = "green", alpha = 0.3
  ) + # manual colors
  geom_line(color = "steelblue", size = 0.8) +
  geom_line(aes(x = date, y = price_o), size = 0.8, color = "gold") +
  geom_line(aes(x = date, y = price_msci), size = 0.8, color = "#332288") +
  geom_line(aes(x = date, y = price_aegean), color = "#88CCEE") +
  geom_line(aes(x = date, y = price_france), color = "#44AA99") +
  geom_line(aes(x = date, y = price_lufthansa), color = "#DDCC77") +
  geom_line(aes(x = date, y = price_ryanair), color = "#CC6677") +
  geom_line(aes(x = date, y = price_tui), color = "#882255") +
  geom_line(aes(x = date, y = price_iceland), color = "#AA4499") +
  geom_line(aes(x = date, y = price_easyjet), color = "#117733") +
  geom_line(aes(x = date, y = price_wizz), color = "#999933") +
  geom_line(aes(x = date, y = price_sas, color = "red")) +
  geom_vline(xintercept = as.Date("2019-09-26"), color = "green4", size = 0.7) + #
indicators
  geom_vline(xintercept = as.Date("2020-02-19"), color = "green4", size = 0.7) +

```

```

geom_vline(xintercept = as.Date("2020-02-26"), color = "red4", size = 0.7) +
  geom_vline(xintercept = as.Date("2020-03-25"), color = "red4", size = 0.7) + # text
  annotate("text", x = as.Date("2019-09-15"), y = 4000, label = "Estimation window", color =
"black", size = 3, angle = 90) +
  annotate("text", x = as.Date("2020-04-02"), y = 4000, label = "Event window", color =
"black", size = 3, angle = 90) +
  labs(title = "Historical European Prices, Figure 3",
    x = "Date",
    y = "Stock Price (NOK)",
    caption = "Investing.com datasets (See references)",
    subtitle = "Log form y-axis") + # titles
  scale_y_continuous(labels = label_number(), expand = c(0, 0), trans = "log10") + # choosing
form for number
  scale_x_continuous(expand = c(0, 0)) +
  theme(axis.text.x=element_text(angle = 0),
    legend.position = "none",
    plot.title = element_text(hjust = 0.5, size = 17)) + # titles
  scale_x_date(date_breaks = "3 month") # tick marks

# making color table
cols <- c(
  "Aegean Airlines" = "#88CCEE",
  "Air France-KLM" = "#44AA99",
  "easyJet" = "#117733",
  "Wizz Air" = "#999933",
  "SAS" = "red",
  "Lufthansa" = "#DDCC77",
  "Ryanair" = "#CC6677",
  "TUI" = "#882255",
  "Iceland air" = "#AA4499",
  "OSEBX" = "gold",
  "MSCI" = "#332288",
  "NAS" = "steelblue"
)

```

```

# dataset
tbl <- tibble(Company = names(cols))

# plotting
ggplot(tbl, aes(x = 1, y = Company, label = Company, fill = Company)) +
  geom_label(color = "white", label.size = 0, label.padding = unit(0.12, "lines")) +
  scale_fill_manual(values = cols, guide = "none") +
  scale_x_continuous(limits = c(0.98, 1.02), expand = c(0, 0)) +
  labs(x = NULL, y = NULL) +
  theme_void()

# first we remove the indexes, then we change all change columns from character and to
decimals
europe_avg <- europe %>%
  select(-price_msci, -open_msci, -high_msci, -low_msci, -vol_msci, -price_o, -open_o, -
high_o, -low_o, -vol_o) %>%
  mutate(
    across(
      starts_with("change_"),
      ~ .x %>%
        stringr::str_replace_all(",", ".") %>%
        readr::parse_number() / 100
    )
  ) %>%
  arrange(desc(date)) %>%
  mutate(avg_ret = rowMeans(across(starts_with("change_")), na.rm = TRUE)) %>%
  filter(date >= as.Date("2019-09-26"),
         date <= as.Date("2020-02-19"))

# plotting
ggplot(europe_avg, aes(x = avg_ret)) + # histogram
  geom_histogram(aes(y = after_stat(density)),
                bins = 10, fill = "steelblue", color = "white") + # density line

```

```

geom_density(linewidth = 0.5) + # titles
labs(title = "Historical average European returns, Figure 5",
      x = "European returns",
      y = "Density",
      caption = "https://www.investing.com/") + # titles
theme(axis.text.x=element_text(angle = 0),
      legend.title=element_blank(),
      legend.position = "none",
      plot.title = element_text(hjust = 0.5, size = 17))

# making dataframe of python results
data3 <- data.frame(
  Window = c("[-10, +10]", "[-5, +5]", "[-3, +3]", "[-1, +1]"),
  n = c(21, 11, 7, 3),
  Std_err = c(0.113, 0.138, 0.069, 0.052),
  CAR = c(-28.45, 9.84, -24.01, -14.16),
  t_stat = c(-0.56, 0.21, -1.37, -1.58)
)

data2 <- data.frame(
  Variable = c("Constant", "OSEBX excess", "R^2", "Observations"),
  Coefficient = c(-0.0005, 1.7795, 0.062, 99),
  Std_err = c(0.004, 0.482, "", ""),
  t_stat = c(-0.137, 3.695, "", "")
)

data1 <- data.frame(
  Window = c("[-10, +10]", "[-5, +5]", "[-3, +3]", "[-1, +1]"),
  n = c(210, 110, 70, 30),
  CAAR = c("-14.97%", "-9.24%", "4.04%", "3.94%"),
  t_stat = c(-3.33, -1.922, 0.940, 1.786)
)

# table

```

```

data1 %>%
  kable("html", caption = "Table 4: Event Study Results for ten European airlines",
        align = c("c", "c", "c", "c"),
        col.names = c("Window", "n", "CAAR", "t-stat")) %>% # styling preference
  kable_styling(
    bootstrap_options = c("striped", "hover", "condensed", "responsive"),
    full_width = FALSE,
    position = "center",
    font_size = 14
  ) %>%
  column_spec(1, bold = TRUE) %>% # colors based on value indicator
  column_spec(3, color = ifelse(grepl("-", data1$CAAR), "red", "green")) %>%
  column_spec(4, color = ifelse(grepl("-", data1$t_stat), "red", "green")) %>%
  add_header_above(c(" " = 1, "Results" = 3)) %>%
  footnote(
    general = "CAAR = Cumulative average abnormal returns for different event windows
around the WHO announcement on 11 March 2020",
    general_title = "Note:",
    footnote_as_chunk = TRUE
  )

# more tables
data3 %>%
  kable("html", caption = "Table 3: Event Study Results for Norwegian Air Shuttle",
        align = c("c", "c", "c", "c", "c"),
        col.names = c("Window", "n", "Std. Err", "CAR", "t-stat")) %>% # styling
  kable_styling(
    bootstrap_options = c("striped", "hover", "condensed", "responsive"),
    full_width = FALSE,
    position = "center",
    font_size = 14
  ) %>%
  column_spec(1, bold = TRUE) %>% # colors based on value indicator

```

```

column_spec(4, color = ifelse(grepl("-", data3$CAR), "red", "green")) %>%
column_spec(3, color = ifelse(grepl("-", data3$Std_err), "red", "green")) %>%
column_spec(5, color = ifelse(grepl("-", data3$t_stat), "red", "green")) %>%
add_header_above(c(" " = 1, "Results" = 4)) %>%
footnote(
  general = "CAR = Cumulative Abnormal Return. t-stat = t-statistic for event window
significance.",
  general_title = "Note:",
  footnote_as_chunk = TRUE
)

# no scientific numbers
data2$Coefficient[1] <- format(as.numeric(data2$Coefficient[1]), scientific = FALSE, nsmall
= 4)

# even more tables
data2 %>% # styling
kable("html", caption = "OLS Regression results for Norwegian Air Shuttle",
  align = c("c", "c", "c", "c"),
  col.names = c("Variable", "Coefficient", "Std.err", "t-stat")) %>%
kable_styling(
  bootstrap_options = c("striped", "hover", "condensed", "responsive"),
  full_width = FALSE,
  position = "center",
  font_size = 14
) %>% # styling
column_spec(1, bold = TRUE) %>%
add_header_above(c(" " = 1, "Results" = 3)) %>%
footnote(
  general = "Estimation window: 26 Sep 2019 - 19 Feb 2020",
  general_title = "Note:",
  footnote_as_chunk = TRUE
)

```


Extra:

Variable	Coefficient	Std. err	t – stat
Constant	-0.0005	0.004	-0.137
OSEBX excess	1.7795	0.482	3.695
R^2	0.062		
Observations	99		

Table 2. OLS regression results for Norwegian Air Shuttle (estimation window: 26 Sep 2019 – 19 Feb 2020)

Window	n	Std err	CAR	t - stat
[-10, +10]	21	0.113	-28.45%	-0.56
[-5, + 5]	11	0.138	+9.84%	+0.21
[-3, + 3]	7	0.069	-24.01%	-1.37
[-1, +1]	3	0.052	-14.16%	-1.58

Table 3. Cumulative abnormal returns for Norwegian Air Shuttle across alternative event windows around the WHO announcement on 11 March 2020.

Window	N	CAAR	t - stat
[-10, +10]	210	-14.97%	-3.233
[-5, +5]	110	-9.24%	-1.992
[-3, +3]	70	4.04%	0.940
[-1, +1]	30	3.94%	1.786

Table 4: Cumulative average abnormal returns (CAAR) across ten European airlines for different event windows around the WHO announcement on 11 March 2020.